

HARSH GUJARATHI

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EDUCATION

Georgia Institute of Technology <i>Master of Science in Computer Science</i>	Atlanta, GA August 2026 – 2028
Birla Institute of Technology and Science, Pilani <i>Bachelor of Engineering in Computer Science (Minor in Data Science); CGPA: 9.2/10.0</i>	Goa, India November 2020 – June 2024

RESEARCH EXPERIENCE

Undergraduate Research Assistant, Confluences of Computer Science Lab <i>Prof. Kunal Korgaonkar, BITS Pilani</i>	January 2023 — July 2024 Goa, India
- Architected an LLM-driven modernization framework to parse complex, undocumented legacy codebases; parsing based on code specifications to understand and refactor legacy systems. - Engineered a privacy-preserving middleware layer atop Zulip; deployed a distributed testbed to implement mixnet-style message batching and encrypted sender identifiers. - Mitigated trace-through traffic analysis in controlled environments and identified critical configuration enforcement vulnerabilities, documenting the tension between cryptographic security and message latency.	
Undergraduate Research Assistant, IoT Research Lab <i>Prof. Neena Goveas, BITS Pilani</i>	August 2021 — July 2024 Goa, India
- Engineered a reproducible Digital Twin for campus solar installations, benchmarking gradient boosting models (XGBoost) against Physics-Informed Neural Networks (PINNs). - Achieved a 10% increase in predictive accuracy with PINNs but identified a 5x penalty in model size, quantifying the critical trade-off between model fidelity and real-time deployment viability. - Modeled departmental TA allocation as an NP-Hard Constraint Satisfaction Problem; led a team of 7 to engineer heuristic scheduling algorithms that optimized assignments under strict course/student constraints. - Developed <i>Divyang</i>, an Android app for prosthetic recommendations; deployed an end-to-end MobileNet SSD model finetuned via TensorFlow Lite to enable efficient inference on resource-constrained edge devices.	

PROFESSIONAL EXPERIENCE

Software Engineer <i>Microsoft — Azure Migrate</i>	July 2024 — July 2026 Hyderabad, India
- Delivered the AI-First experience for Azure Migrate Agent by integrating RAG with a self-built NL2KQL tool for custom KQL query generation, improving agent accuracy from 40% to 85%. - Owned the IIServer discovery and assessment microservice end-to-end — a distributed, workflow-driven service built on a pub/sub model with concurrent task orchestration; extended native coverage to .NET Core, Python, PHP, and Java apps across VMWare vCenter environments. - Spearheaded the re-architecture of the compound assessment engine to integrate event-driven signals from Azure Resource Graph, enabling a more extensible and scalable assessment pipeline. - Drove expansion of daily E2E test coverage across all 44 Azure regions, safeguarding service reliability and data integrity for global enterprise-scale deployments. - Sole owner of a critical fallback extension for non-container workloads; architected and shipped the feature, driving assessment coverage to 99.9% across 100K+ enterprise workloads. - On-call rotation across a Tier-1 global service; executed end-to-end RCAs and coordinated cross-team mitigations, reducing Sev-2 incidents and sustaining the microservice at a 99.9% SLA.	
Member of Technical Staff Intern <i>DevRev</i>	January 2024 — June 2024 Bengaluru, India
- Strengthened the reliability of the Global Notifications microservice handling 60K+ daily events, spanning rate limiting, throttling, and deduplication. - Engineered distributed rate limiting via Redis timed buckets with event-based throttling, eliminating notification drops under bursty load. - Built a fault-tolerant deduplication engine reducing spam by 85%; implemented hierarchical clustering with $O(\log n)$ lookup via dendrogram traversal for topic-aware redundancy removal.	

- Engineered an LLM-driven summarization pipeline (GPT-4 via LangChain) for digest generation, compressing data throughput by **6x** and improving actionable signal density.
- Drove performance profiling across high-throughput flows (reminders, tasks), surfacing bottlenecks and improving system reliability for global delivery.

Software Engineer Intern

May 2023 — July 2023

Microsoft — Azure Backup and Site Recovery

Hyderabad, India

- Instrumented customer-side meter telemetry for the Azure Business Continuity Center using **OpenTelemetry** and **Geneva**, enabling real-time observability for V1 services.
- Packaged the telemetry instrumentation as a reusable **NuGet library** with **99% test coverage**, adopted by adjacent engineering teams to accelerate integration for continuity scenarios.

SELECTED PROJECTS

IP-Wrapper

October 2022

Developer

github.com/Neo-PL/IP-Wrapper

- Developed a **multithreaded image processing engine** in C++ implementing custom convolution algorithms for operations like Gaussian blur, Sobel edge detection, and sharpening.
- Optimized performance using **OpenMP parallelization** to distribute channel processing and pixel computations, achieving efficient execution on multi-core architectures.
- Designed a modular **kernel-based architecture** allowing dynamic loading of custom filter masks and support for both monochromatic and RGB image streams.
- **Stack:** C++, OpenCV, OpenMP, CMake, Parallel Computing.

Amplitude-Phase Recombination (APR)

April 2023

Deep Learning Developer

github.com/alphaNewrex/APR-Jupyter

- Developed a novel **frequency-domain data augmentation** pipeline in **PyTorch** that synthesizes diverse training samples by recombining the amplitude and phase spectra of images via FFT/IFFT operations.
- Integrated the APR algorithm with state-of-the-art CNN architectures (**DenseNet**, **ResNet**, **WideResNet**) to enhance feature learning and model generalization on the CIFAR-10/100 datasets.
- Conducted extensive robustness benchmarking using the **CIFAR-10-C** dataset, demonstrating improved resilience against 15+ types of common image corruptions compared to standard augmentation baselines.
- **Stack:** Python, PyTorch, NumPy, Computer Vision, Fourier Transforms, Deep Learning.

TECHNICAL SKILLS

Languages

Python, C++, Java, C#, Go, SQL, TypeScript, JavaScript

ML & Deep Learning

PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, RAG, LLM Agents, CNNs, FFT/IFFT

Distributed Systems

Kubernetes, Docker, Kafka, Pub/Sub, gRPC, Azure Resource Graph, Open Telemetry

Data & Backend

PostgreSQL, MongoDB, Redis, .NET Core, Node.js, NumPy, Pandas

TEACHING EXPERIENCE

Teaching Assistant, Department of CSIS

August 2022 — Dec 2023

BITS Pilani

Goa, India

- **Courses:** Operating Systems, Computer Architecture (x2), Software for Embedded Systems, Data Structures & Algorithms, Object Oriented Programming, Logic in Computer Science.
- Facilitated lab sessions and tutorials for **7 undergraduate and graduate courses**, mentoring students on complex topics including **Linux system calls**, **Verilog/FPGA design**, and **MIPS architecture**.
- Designed comprehensive **lab questions and reference materials** for core systems courses, and evaluated assignments for graduate-level VLSI architectures.

Instructor, Centre for Technical Education

March 2022 — June 2022

BITS Pilani

Goa, India

- Designed the curriculum and taught a comprehensive **Android Development** course to a batch of 45 students, covering application lifecycle, UI/UX design, and backend integration.